16. Find the unit vector of $\mathbf{A} = 4\mathbf{i} - 3\mathbf{j}$.
    A. $\frac{4}{5}i + \frac{3}{5}j$
    B. $\frac{4}{5}i - \frac{3}{5}j$
    C. $\frac{4}{5}i - \frac{3}{5}j$
    D. $\frac{4}{25}i + \frac{3}{25}j$

17. What is the scalar product of the vectors.
    $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$?
    A. -1
    B. -3
    C. 3
    D. 7

18. The change in momentum of a body also define its _____ .
    A. force
    B. power
    C. impulse
    D. potential energy

19. A boy drags a load on a horizontal smooth floor with a force of 4N applied at an inclination of 60° to the horizontal. If the load moves through a distance of 3m. What is the work done by the force?
    A. 4 J
    B. 6 J
    C. 8 J
    D. 10 J

20. How much time will be required to perform 520 J of work at a rate of 20 W?
    A. 26 s
    B. 24 s
    C. 20 s
    D. 16 s

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EXAM CODE 01

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COLLEGE OF DISTANCE EDUCATION
SEMESTER I QUIZ I, 2020/2021
BACHELOR IN MATHEMATICS / SCIENCE EDUCATION
4 YR BME/BSE YR 1
PHY 103: GENERAL PHYSICS I
EXAM CODE: 01

NAME: .....

REGISTRATION NO.:.....SIGNATURE:.....

STUDY CENTRE:.....TIME ALLOWED: 30 MINUTES

INSTRUCTION: *Answer all questions by circling the correct option in ink.*

1. In freefall, an object is seen to have a (an) _____ .
    A. constant velocity
    B. constant acceleration
    C. increasing acceleration
    D. decreasing acceleration

2. An object moving in a particular direction is described by its _____.
    A. speed
    B. velocity
    C. acceleration
    D. none of the above

3. If the velocity of a body is doubled, its kinetic energy _____ .
    A. gets doubled
    B. does not change
    C. is halved
    D. increases four-fold

4. The area under the velocity–time graph represents _____ .
    A. acceleration
    B. average velocity
    C. distance
    D. average acceleration

5. Power can be expressed in _____ .
   A. N/m
   B. Nm/s
   C. Nm^2/s^2
   D. Kgm/s

6. According to the definition of mechanical work, pushing on a rock accomplishes no work unless there is _____ .
   A. a net force
   B. movement
   C. an opposing force
   D. movement in the same direction as the direction of the force

7. Find the work done in moving an object along a vector $\mathbf{r} = 3\,\mathbf{i} + 2\,\mathbf{j} - 5\mathbf{k}$ If the applied force is $\mathbf{F} = 2\,\mathbf{i} - \mathbf{j} - \mathbf{k}$.
   A. 2 J
   B. 3 J
   C. 9 J
   D. 13 J

8. If $\mathbf{A} \times \mathbf{B} = 0$ and if **A** and **B** are non- zero vectors, then _____ .
   A. **A** is parallel to **B**
   B. **A** is perpendicular to **B**
   C. Angle between **A** and **B** is 270°
   D. Angle between **A** and **B** is 45°

9. A process of comparing a property to a well-defined and agreed –upon reference is called _____ .
   A. Referral
   B. Measurement
   C. Magnitude
   D. Comparison

10. A load causes a wire to stretch by x. If the same load is applied to another wire of the same material but twice as long and with twice the diameter. Find the extension?
    A. 2x
    B. x

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    C. $x/2$
    D. $x/4$

11. If $M^x L^y Y^{-z}$ is the dimensional formula for work, find the value of $3x + 2y + z$.
    A. 4
    B. 5
    C. 6
    D. 7

12. The stress on a wire that supports a load depends on the following **except** _____ .
    A. diameter of the wire
    B. acceleration due to gravity
    C. mass of the load
    D. length of the wire

13. In an inelastic collision _____ is/are conserved.
    A. kinetic energy but not momentum
    B. momentum but not kinetic energy
    C. neither momentum nor kinetic energy
    D. both momentum and kinetic energy

14. If **i, j** and **k** are unit vectors in the x,y,z directions respectively then the **cross product** between **j** and **i** (ie **j** x **i**) is
    A. i
    B. j
    C. –k
    D. k

15. If vector **B** is added to vector **A** the result is $6\mathbf{i} + \mathbf{j}$. If **B** is subtracted from **A** the result is $-4\mathbf{i} + 7\,\mathbf{j}$. what is the magnitude of **B**?
    A. 4.46
    B. 5.83
    C. 6.72
    D. 6.94